

Explanation-Supervised Attention for Multi-Label Thoracic Disease Classification: A Variant CNN Trained from Weak Localization Labels

Israel Agyekum `iagyekum006@st.ug.edu.gh`, Joel Dadi-Klutse
`jdadi-klutse@st.ug.edu.gh`, and Eric Okyere `eokyerel46@st.ug.edu.gh`

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Proposal

1. Problem and motivation. Thoracic diseases (pneumonia, edema, fibrosis, emphysema, pneumothorax) are leading causes of global morbidity and mortality, and chest radiography is the most accessible imaging modality—yet accurate reading depends on scarce radiological expertise, causing diagnostic delay and inter-reader inconsistency, especially in resource-limited settings such as much of sub-Saharan Africa. CNNs match expert performance here but behave as opaque “black boxes,” and the usual remedy—post-hoc Grad-CAM—is computed *after* training and is not optimized to be correct, so its heatmaps can be unfaithful and clinically misleading. We address the *algorithmic* gap: rather than explaining a fixed model after the fact, we design a model that is *trained to attend to the right regions*. The novelty is methodological—a new attention mechanism and objective—not the mere application of existing tools.

2. Background and reading. We will review transfer-learning backbones—ResNet [1], DenseNet [2], EfficientNet [3]—and the CheXNet benchmark [4]; the ChestX-ray14 dataset and its weakly-supervised localization baseline [5]; attention, especially CBAM [6]; explanation methods, especially Grad-CAM [7]; imbalance-aware training via focal loss [8]; label-dependency modelling for multi-label recognition [9]; and explanation/attention supervision (“right for the right reasons”) [10]. We cover both predictive performance and the open problem of *faithful*, trainable interpretability.

3. Data. We use NIH ChestX-ray14: 112,120 frontal images from 30,805 patients with 14 thoracic findings; because images carry several findings at once we treat the task as *multi-label*. Crucially, a clinician-annotated subset provides ≈ 984 bounding boxes over 880 images for 8 findings—this weak localization signal is what makes our algorithmic contribution possible. We use a class-balanced subset for feasibility, split strictly at the *patient level* to prevent the leakage that occurs when one patient’s images span train and test, and apply resizing, normalization and augmentation.

4. Method (algorithmic core). Starting from ImageNet-pretrained ResNet50/DenseNet121/EfficientNetB0 (timm/torchvision) as *baselines*, we propose a *variant* model with two new components. (i) *Explanation-supervised attention*: we redesign the spatial-attention branch so its map is a first-class, trainable output, supervised by the bounding-box masks through an alignment loss $\mathcal{L}_{\text{attn}}$ (Dice/MSE on boxed images) and regularized by sparsity/entropy where no box exists—turning explanation from post-hoc into part of optimization. (ii) *Label-correlation-aware objective*: instead of treating the 14 labels independently, we add a learned co-occurrence/graph term so predictions respect clinical label dependencies, combined with focal/class-weighted BCE for imbalance. The total loss is $\mathcal{L} = \mathcal{L}_{\text{cls}} + \lambda_1 \mathcal{L}_{\text{attn}} + \lambda_2 \mathcal{L}_{\text{corr}}$. We adapt published CBAM/Grad-CAM/timm code, but the supervised-attention module and correlation-aware loss are our own design, yielding a new architecture rather than a plug-in.

5. Evaluation. *Classification*: per-class and macro Precision, Recall, F1 and ROC-AUC (the standard ChestX-ray14 metric), with confusion matrices and DeLong’s test for AUC significance. *Localization (decisive for the algorithmic claim)*: we *quantify* explanation quality with IoU and pointing-game accuracy against the bounding boxes, comparing our supervised attention to vanilla Grad-CAM. Controlled ablations isolate each new component ($\mathcal{L}_{\text{attn}}$, $\mathcal{L}_{\text{corr}}$) and each backbone. Qualitatively we expect per-label ROC/training curves and heatmap overlays. The central question: does explanation supervision improve localization faithfulness while keeping AUC competitive and adding *no* inference cost?

6. Scope and unique contribution. Standalone project for CSCD 618/DSCD 604—not combined with any other course or research. The unique, *algorithmic* contribution is a CNN variant with an explanation-supervised attention module and a label-correlation-aware objective, evaluated for *measured* localization faithfulness rather than visual plausibility, with potential for publishable work.

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